

WEEK 1

Introduction to the Pharmaceuticals Landscape & Terminology

Why this industry is structurally different — and who gets paid for what.

Andi Cupallari · Applied Data Science and AI in Pharma Commercial · Fall 2026 · Monday, September 14

Where we're going tonight

2.5 hours · discussion-heavy · one short notebook tour · one in-class exercise

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Welcome & course logistics

Grading, term project, mentors, AI-use policy — 20 min

1

Why pharma commercial is structurally different

Chooser ≠ payer ≠ consumer — 15 min

2

The stakeholder map

Manufacturer, wholesaler, pharmacy, HCP/HCO/IDN, payer, PBM, GPO, patient, policymaker — 30 min

3

Follow the pill, follow the money

Script → fill → claim → revenue; NRx / TRx / NBRx — 20 min

4

Gross-to-net and the brand P&L

Why list price is a fiction; reading GLUCOREN's P&L — 20 min

5

Codes & acronyms

ICD-10, NDC, NPI, HCPCS — the keys that join every dataset — 10 min

6

Product lifecycle & launch archetypes

Launch → in-line → LOE; IQVIA's four archetypes — 15 min

7

AI and agentic AI in commercial — why now

What leaders say they're deploying, and where it fails — 15 min

8

Term project use cases, claims tour, exercise

Team formation · notebook tour · stakeholder exercise — 30 min

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Welcome & course logistics

What this course is for, how you'll be graded, and how the term project works.

Three things you'll leave with in December

From the syllabus — the course has two deliberate goals: fluency and exposure

1 Interview-ready fluency

- ▶ Speak credibly about claims data, MMM, HCP segmentation, omnichannel orchestration, forecasting, NLP and agentic AI
- ▶ Know the vocabulary hiring managers at manufacturers, consultancies and analytics vendors actually use
- ▶ Weeks 1–2 build the domain context; no prior pharma knowledge assumed

2 End-to-end projects you can show

- ▶ Team term project (3–4 people) on a realistic commercial problem: framing → data → model → recommendation
- ▶ Two homeworks (MMM; forecasting) built on synthetic pharma datasets
- ▶ A repo + executive memo a hiring manager can read

3 Sustained industry exposure

- ▶ Guest lectures from senior practitioners with real case studies
- ▶ Every team paired with an industry mentor for the whole semester (introduced Sep 21)
- ▶ Final presentations judged by mentors and an industry panel

Prerequisites: intro stats + ML, Python (pandas, scikit-learn, Jupyter). Everything else we build here.

How you're graded, and the dates that matter

Homework is take-home, due two weeks after release. Late submissions are not accepted.

Component	Weight
Guest-speaker engagement (3 × 5%): questions in advance + 1-page summary	15%
Homework 1 (MMM) and Homework 2 (forecasting)	20%
Midterm exam (in class, Nov 9)	25%
Term project (mentor-supervised)	40%

Office hours: Mondays 5:00–6:45 PM, before class.

Readings are free or distributed via the course site; no textbook to buy.

Grade scale: A ≥ 93, A- ≥ 90, B+ ≥ 87, B ≥ 83 ... F < 60 (full table in syllabus).

Date	Milestone
Sep 14	Week 1 — tonight
Sep 21	Week 2 — claims analytics · mentors introduced · teams formed
Sep 28	Week 3 — MMM · Homework 1 released (due Oct 12)
Oct 19	Week 6 — Forecasting · Homework 2 released (due Nov 2)
Nov 2	Academic holiday — no class
Nov 9	Midterm exam
Nov 16 / Nov 30	Guest speakers 1 & 2 (agentic AI; the journey of MMM)
Nov 23 / Dec 7	Term project progress check-ins #1 and #2
Dec 14	Industry panel + final presentations (20 min + 5 Q&A)

The term project: mentor-supervised, semester-long, 40% of the grade

Teams of 3–4 • a realistic pharma commercial problem of your choosing • end-to-end analytical solution



Your mentor (a senior practitioner)

- ▶ Shapes the problem so it reflects an authentic commercial question
- ▶ Sanity-checks your data plan, modeling choices, stakeholder framing
- ▶ Surfaces what textbooks miss: regulatory friction, org politics, market dynamics
- ▶ Does NOT write code or do the analysis — you own scheduling, agendas, follow-ups
- ▶ Cadence: 30–45 min every two weeks (~6 touchpoints)

Deliverables (Dec 14)

- ▶ 20-minute in-class presentation + 5 min Q&A, mentor and panel in the room
- ▶ A code repository (notebooks + README) any teammate could clone and re-run
- ▶ A 2–4 page executive memo: problem, approach, results, recommendation, limitations
- ▶ A short mentor-acknowledgment paragraph

Use-case catalog handed out tonight (Week_01_Term_Project_Use_Case_Catalog) — 15 starting points beyond the five archetypes in the syllabus.

An applied-AI course: use AI tools thoughtfully, with disclosure

Applies to ChatGPT, Claude, Copilot, Gemini and similar. Default answer when unsure: “yes, with disclosure.”

Permitted

- ▶ Brainstorming, debugging, code completion
- ▶ Explaining unfamiliar concepts, summarizing readings
- ▶ Drafting boilerplate, generating synthetic data
- ▶ Accelerating exploratory analysis

Required

- ▶ A short “AI use” appendix on any non-trivial AI-assisted work: tool, prompts (or a representative sample), what you changed
- ▶ One paragraph is fine
- ▶ You are responsible for everything you submit — models hallucinate citations, statistics, regulatory facts and clinical claims

Not permitted

- ▶ Submitting AI text/code as your own without review and substantive editing
- ▶ Using AI to bypass the learning objective of an assignment
- ▶ Pasting confidential, proprietary or patient-identifiable data into an external tool
- ▶ AI on the midterm or in-class assessments unless stated

Industry reality check: the same rules apply on the job. A 2025 IQVIA survey of 107 commercial leaders found privacy/regulatory constraints (51%) are the #1 barrier to scaling AI in pharma commercial.

01

Why pharma commercial is structurally different

Most analytics intuition comes from B2C. Much of it breaks here.

The person who chooses doesn't pay, and the person who pays doesn't consume

HCP

Chooses

The physician (or NP/PA) decides what to prescribe. Influenced by evidence, guidelines, habit, peers, samples, reps — and by what the payer will cover.

Payer / PBM

Pays (mostly)

Insurer, employer, Medicare/Medicaid — and the PBM that manages the pharmacy benefit. Decides formulary tier, prior authorization, copay. Wants rebates and low net cost.

Patient

Consumes

Takes (or doesn't take) the drug. Pays a copay or coinsurance. Rarely sees the list price. Adherence, persistence and affordability are the patient-side levers.

Contrast with B2C: one person browses, decides, pays and uses — and the retailer sees the whole transaction in first-party data. In pharma the manufacturer sees none of it directly. It buys back a projected view of its own market from data vendors. That single fact explains most of Week 2.

Six things that break your B2C intuition

Each one shows up again as a modeling problem later in the course

Structural feature	What it means for analytics	Where it bites (week)
No first-party transaction data	You license syndicated Rx/claims panels and projections; sample bias and patient capture are your problem	Wk 2 datasets • Wk 6 forecasting
Decision maker ≠ payer ≠ consumer	Three targets, three data types, three behavioral models; “conversion” is a chain, not an event	Wk 4 segmentation • Wk 5 NBA
Access gates on every script	Formulary tier, prior auth, step edits and copay determine whether an Rx becomes a fill	Wk 2 claims • Wk 3 MMM covariates
Price is not one number	WAC → wholesaler → rebates → 340B/Medicaid → copay cards → net. Revenue ≠ volume × list	Tonight • Wk 3 ROI
Regulated promotion	FDA-approved label only; Sunshine Act reporting; no direct incentives to HCPs; DTC rules	Wk 5 omnichannel • Wk 7 NLP
A lifecycle with a cliff	Patent expiry / loss of exclusivity (LOE) can erase 80–90% of brand volume in a year	Wk 6 forecasting

A few numbers to calibrate the stakes

All figures from IQVIA white papers in this week's reading pack

\$3.1B

estimated cost to bring a new therapy to market; 11% composite success rate from Phase 1 to submission

50–70%

of launch brands underperform their first-year forecast (IQVIA study of ~12,000 launches, 8 countries)

\$179.5B

340B program purchases at list price in 2025, +20.3% YoY — grew 3× faster than the rest of the U.S. drug market since 2018

44%

of life-sciences commercial organizations now dedicate >20% of their commercial budget to AI (IQVIA 2025 survey, n=107)

- ▶ **Why these matter tonight:** development cost + failure rate explain why commercial teams obsess over every launch; the forecast-miss statistic is the single best argument for the analytics you'll build in this course.
- ▶ **340B** is a federal program giving Medicaid-like discounts to qualifying hospitals and clinics. It is one of the largest wedges between list and net price — and a live policy fight (a rebate-model pilot began with 10 drugs in 2026).
- ▶ **AI spend** tells you where the jobs are: 58% of leaders report $\geq 2\times$ one-year ROI on AI initiatives, yet only 11% say their data is “fully sufficient” to support AI. Data readiness is the bottleneck — and the opportunity.

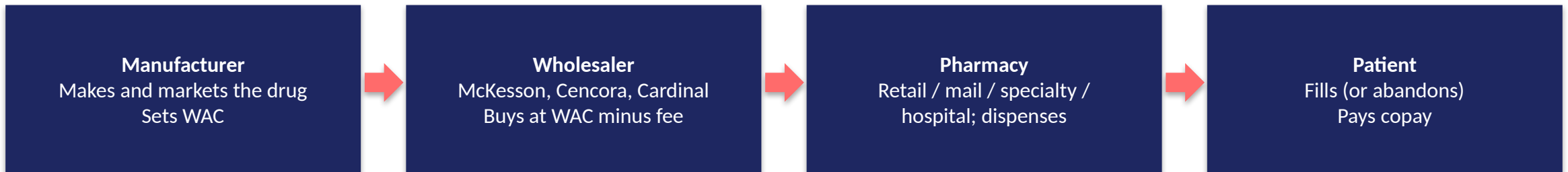
02

The stakeholder map

Nine players. Two flows — product and money — that run in opposite directions.

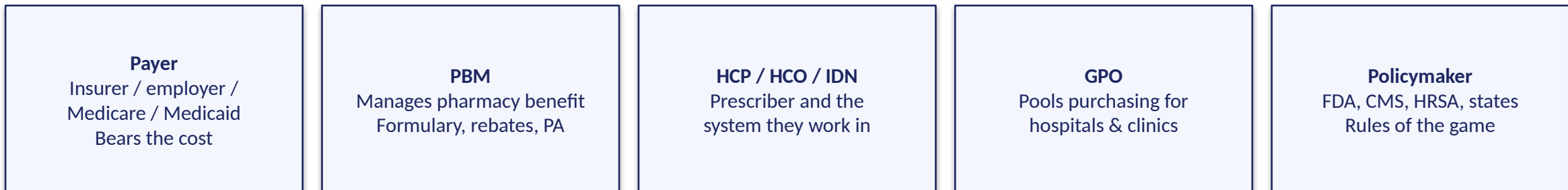
The U.S. pharmaceutical ecosystem on one slide

Top row = product moves • bottom row = money and influence



▲ *Product flow (left → right) — tracked by shipment data (DDD-style)*

▼ *Money & influence — tracked by claims, rebates, contracts*



► **Follow the money:** Payer pays PBM/pharmacy → pharmacy paid per fill → wholesaler pays manufacturer WAC minus fees → manufacturer pays rebates back to PBM/payer, chargebacks to wholesalers, and 340B/Medicaid discounts.

► **Follow the data:** every hand-off leaves a record with a code on it (NDC, NPI, ICD-10, plan ID). Those codes are how Week 2's datasets join together.

Manufacturer and wholesaler

Manufacturer

Pfizer, Novo Nordisk, an emerging biotech — or our fictional Aspen Therapeutics (maker of GLUCOREN). Functions: R&D, medical affairs, commercial (brand, sales, market access, insights/analytics).

WHAT THEY WANT

Maximize net revenue over the exclusivity window; get on formulary at a good tier; grow share of new-to-brand patients; defend against LOE. Constrained by label, compliance, and a 12–15-year patent clock.

DATA THEY GENERATE

Shipments (ex-factory sales), CRM/field activity (calls, samples), promotional spend, rebate contracts, patient-support-program enrollments, MSL insights.

WHERE ANALYTICS TOUCHES THEM

Almost everything in this course: MMM (Wk 3), segmentation (Wk 4), NBA (Wk 5), forecasting (Wk 6), NLP on field notes (Wk 7), agentic workflows.

Wholesaler / distributor

Three firms (McKesson, Cencora, Cardinal Health) move >90% of U.S. drug volume from manufacturer to pharmacy and hospital.

WHAT THEY WANT

Thin margins on huge volume: fee-for-service from manufacturers, prompt-pay discounts, spread on generics. Wants inventory efficiency and predictable demand.

DATA THEY GENERATE

Sales-to-outlet data (the source of DDD-style datasets), chargeback claims when a GPO/340B contract price applies, inventory levels.

WHERE ANALYTICS TOUCHES THEM

Demand/supply forecasting; channel inventory “pipeline” effects that distort launch curves; chargeback reconciliation in gross-to-net.

Payer and PBM — the access gate

Payer

Whoever bears the cost: commercial insurers (UnitedHealthcare, Elevance, Aetna...), employers who self-insure, Medicare (Parts B and D), Medicaid, the VA.

WHAT THEY WANT

Lowest net cost for acceptable outcomes; predictable budgets; avoid paying for drugs used off-guideline. Increasingly demands real-world evidence of value.

DATA THEY GENERATE

Adjudicated medical and pharmacy claims, enrollment/eligibility, prior-authorization decisions, formulary documents, HEOR value dossiers submitted to them.

WHERE ANALYTICS TOUCHES THEM

Payer/market-access analytics: formulary status impact, contract simulation, evidence generation; payer covariates in MMM and forecasting.

PBM (Pharmacy Benefit Manager)

CVS Caremark, Express Scripts (Cigna), Optum Rx (UnitedHealth) manage ~80% of U.S. prescriptions on behalf of payers.

WHAT THEY WANT

Rebates from manufacturers in exchange for formulary position; spread and fees; steer volume to owned mail/specialty pharmacies. Their incentive is not simply “lowest price.”

DATA THEY GENERATE

The pharmacy claim itself (adjudicated in real time: covered? tier? copay? PA?), rebate invoices, formulary and utilization-management rules.

WHERE ANALYTICS TOUCHES THEM

Gross-to-net modeling; formulary-change event studies; the “access” covariates that separate promotion effects from coverage effects.

Providers and GPOs — where the prescribing happens

HCP • HCO • IDN

HCP: the individual prescriber (MD, DO, NP, PA), identified by NPI. HCO: the organization (practice, hospital). IDN: an integrated delivery network — a system that owns hospitals, clinics, often a health plan (Kaiser, Intermountain).

WHAT THEY WANT

Best outcome for the patient in the time available; minimal admin friction (prior auth is hated); professional reputation; for IDNs, system-wide cost and protocol adherence — decisions move from individual to committee.

DATA THEY GENERATE

Every claim carries the prescriber's NPI; EHR records (diagnosis, labs, notes); field-call feedback; conference attendance; publications (for KOLs).

WHERE ANALYTICS TOUCHES THEM

HCP segmentation and targeting (Wk 4), NBA and channel fatigue (Wk 5), KOL identification, account-based analytics for IDNs, NLP on field notes (Wk 7).

GPO (Group Purchasing Organization)

Vizient, Premier, HealthTrust negotiate purchasing contracts for member hospitals and clinics; relevant mainly for physician-administered and hospital drugs.

WHAT THEY WANT

Volume-based discounts and administrative fees; member hospitals want lower acquisition cost for drugs they buy and bill (the “buy-and-bill” model).

DATA THEY GENERATE

Contract prices that trigger wholesaler chargebacks; membership rosters; hospital purchasing data.

WHERE ANALYTICS TOUCHES THEM

Channel and contract analytics for infused/injected products (oncology, immunology); a driver of gross-to-net for hospital-administered brands.

Patients and policymakers — the fourth “P”

Patient

The consumer, and increasingly a decision-maker: DTC advertising, copay cards, patient-support programs, online communities. Identity is protected (HIPAA) — you work with de-identified tokens.

WHAT THEY WANT

Feel better, afford the drug, avoid hassle. Out-of-pocket cost is the #1 driver of prescription abandonment; complexity of dosing drives non-adherence.

DATA THEY GENERATE

The claims trail (dx, procedures, fills, refills), patient-support-program data, social listening, adherence/persistence measures (PDC, MPR), surveys.

WHERE ANALYTICS TOUCHES THEM

Patient-finding and journey models, adherence prediction, DTC in MMM, social-listening NLP (Wk 7), and privacy/re-identification risk (Wk 2).

Policymaker / regulator

FDA (approval, label, promotion rules), CMS (Medicare/Medicaid; now negotiating Part D prices under the IRA), HRSA (340B), FTC, state legislatures (PBM transparency, contract-pharmacy mandates).

WHAT THEY WANT

Safety and efficacy; affordability and budget control; program integrity; political accountability. Policy changes reprice entire markets overnight.

DATA THEY GENERATE

Approval and label documents, Sunshine Act (Open Payments) disclosures, negotiated “maximum fair price” lists, 340B covered-entity registries.

WHERE ANALYTICS TOUCHES THEM

Scenario analysis in forecasting (Wk 6); compliance constraints on promotion models; policy shocks as structural breaks in time series.

Four tensions you will end up modeling

Every one of these is an analytics project inside a real company

1 Rebates vs. list price

PBMs earn from rebates, which are a percentage of list price — so a higher list price with a bigger rebate can be preferred over a lower list price. Patients on coinsurance pay on the list price. Result: rising list, flat-to-falling net, and a “gross-to-net bubble.”

2 Formulary tier vs. patient copay

The payer wants to steer to the preferred brand; the manufacturer of the non-preferred brand counters with copay cards that neutralize the copay difference. Payers respond with accumulator/maximizer programs. An arms race visible in claims data.

3 340B discounts vs. manufacturer net

Hospitals and clinics buy at deep discounts and often bill payers at full reimbursement. 340B purchases grew 233% (2018–2025) vs 71% for the rest of the market. Manufacturers push a rebate model instead of upfront discounts; hospitals push back on cash-flow grounds.

4 HCP attention vs. promotional pressure

Reps, emails, ads and MSLs all want the same 8 minutes of a physician's day. Over-contact produces fatigue and channel opt-outs — and mis-attributed “response” in naive models. This is the core of Week 5 (Next Best Action).

03

Follow the pill, follow the money

From a diagnosis to a line on the manufacturer's income statement — and the data each step leaves behind.

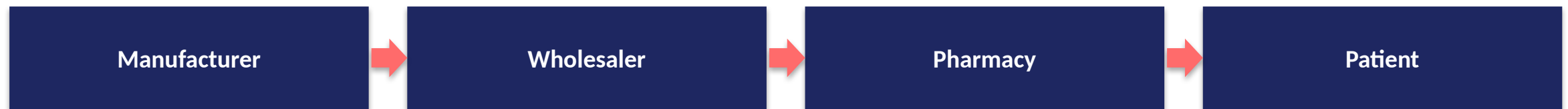
The life of a prescription: script → fill → claim → revenue



Key insight: the manufacturer's revenue is recognized at step 6 when the wholesaler buys — before the patient exists. Demand (steps 1–5) and revenue (step 6) are measured in different systems, at different times, by different companies. Reconciling them is a full-time job.

Product goes one way, money the other — plus a loop of rebates

Simplified U.S. retail brand; hospital “buy-and-bill” drugs add GPOs and chargebacks



Product: ex-factory → distribution center → dispensed. Manufacturer sells at WAC minus a distribution-service fee (~2–4%).

- ▶ **Patient → Pharmacy** — copay / coinsurance at the counter (often offset by a manufacturer copay card)
- ▶ **Payer → PBM → Pharmacy** — reimbursement for the dispensed claim (ingredient cost + dispensing fee), minus PBM spread
- ▶ **Manufacturer → PBM / Payer** — rebates, paid quarterly in arrears, in return for formulary position — the biggest gross-to-net item for many brands
- ▶ **Manufacturer → Wholesaler** — chargebacks when a contract price (GPO, 340B, VA) is lower than WAC; also fees, prompt-pay discounts, returns
- ▶ **Manufacturer → Government** — statutory Medicaid rebates (basic + inflation penalty); Medicare Part D coverage-gap/manufacturer discount; 340B ceiling prices

Rebate loop: manufacturer → PBM → payer (paid quarterly, in arrears)



WHY ANALYSTS CARE

Volume (TRx) lives in pharmacy claims. Revenue lives in finance. The bridge — gross-to-net — is negotiated brand by brand, payer by payer, and is confidential. When you compute “ROI” in Week 3 you will be handed a net contribution per TRx by finance. Never invent it.

The prescription metrics you'll hear in every meeting

Definitions matter: two analysts reporting “share” can mean four different things

Metric	Definition	Used for
NRx	New prescriptions: a script that is not a refill (patient may have used the brand before)	Leading indicator of prescribing behavior; response variable for promotion models
TRx	Total prescriptions = NRx + refills, in a period; often converted to “normalized” 30-day equivalents	Volume, revenue proxy; MMM response variable (Wk 3)
NBRx	New-to-brand: patient had no claim for this brand in the prior 12 months (from longitudinal patient data)	The cleanest measure of share gain at launch; harder to get
Share (TRx / NRx / NBRx)	Brand ÷ market. Market definition (which molecules? which line of therapy?) changes the answer	Competitive tracking; forecast anchoring
Persistence • adherence	Persistence: time on therapy before a gap; adherence: PDC or MPR — proportion of days covered	Patient-journey analytics; forecasting the “refill tail”
Writers • reach • frequency	Prescribers with ≥1 Rx; % of targets contacted; contacts per target per period	Field-force sizing, NBA, fatigue management (Wk 5)

Source-dependent: an “NRx” from a retail panel (projected from ~90% of retail pharmacies) is not the same number as one from a longitudinal claims database with partial payer coverage.

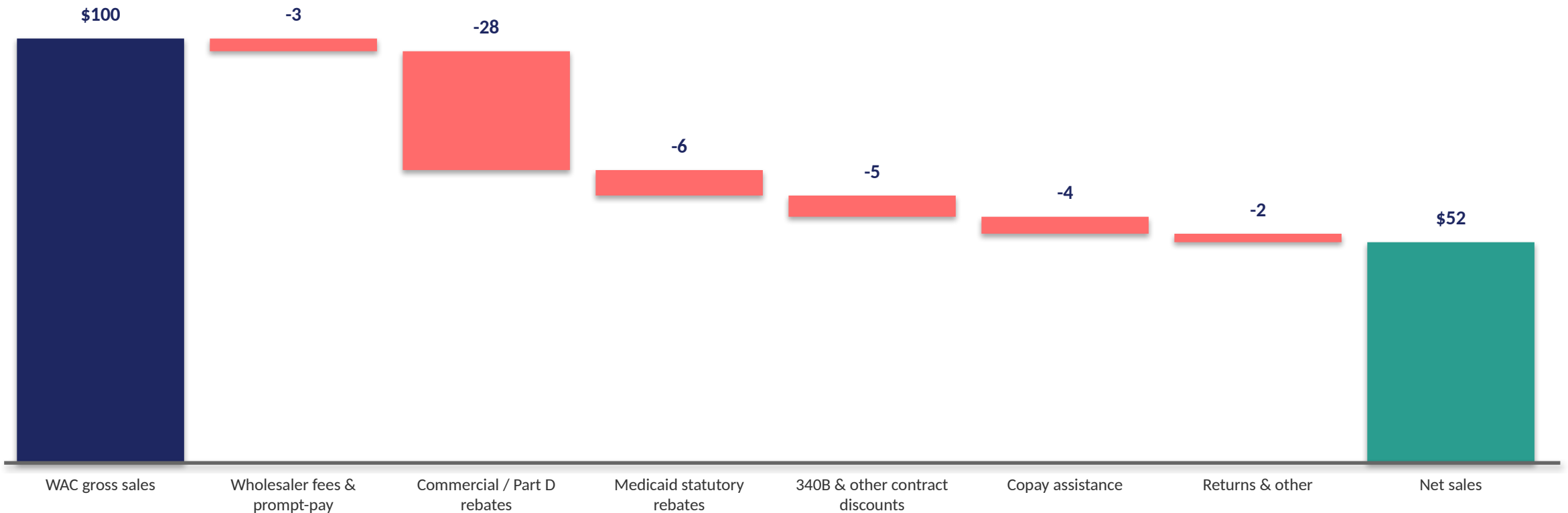
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Gross-to-net and the brand P&L

List price is a fiction. Net price is a negotiation. The P&L is where both meet.

From \$100 of list price to ~\$52 of net: the gross-to-net waterfall

Illustrative for GLUCOREN, a branded oral type-2 diabetes agent with heavy PBM rebating. Real GTN ranges from ~15% (rare disease) to >70% (insulins).



Reading GLUCOREN's brand P&L (illustrative, \$M, year 3 post-launch)

This is what a brand lead looks at monthly. Learn the lines; the exercise tonight asks you to compute the missing ones.

Line	\$M	% of net	What an analyst asks
Gross sales (WAC × units)	1,250	—	Volume × list. Reconciles to wholesaler purchases, not to TRx.
Gross-to-net deductions	(600)	—	48% GTN. Trend? Mix shift toward Medicaid/340B? Rebate rate creep?
Net sales	650	100%	The number the street sees. Net price per TRx = 650 / TRx.
Cost of goods sold	(52)	8%	Small for small molecules; 15–30% for biologics/cell therapy.
Gross margin	598	92%	
Promotion: field force (detailing, samples)	(95)	15%	Reps × cost per rep. Is reach/frequency worth it? → MMM, NBA
Promotion: non-personal (digital, email, conferences)	(30)	5%	Cheapest reach; measured by MMM and experiments
Promotion: DTC (TV, digital consumer)	(45)	7%	Big, lumpy, contested ROI; drives patient requests
Medical affairs (MSLs, publications, HEOR)	(25)	4%	Non-promotional by law; value is evidence and KOL trust
Market access & patient services (copay program ops, hubs)	(20)	3%	Grows with access friction
Brand contribution	383	59%	What the brand hands to the company before overhead and R&D.

Five questions an analyst asks of any brand P&L

You will be asked at least one of these in an interview

1

Where is GTN going?

A 48% GTN today that was 40% three years ago says payers are winning. Decompose by channel (commercial, Part D, Medicaid, 340B) and by rebate rate vs. mix.

2

What is promotional intensity?

Total promotion ÷ net sales. 25–30% is typical for a growing primary-care brand; <10% for a mature or specialty brand. Compare with the launch curve.

3

What is a TRx worth?

Net sales ÷ TRx (normalized) gives net revenue per script; subtract COGS for contribution per script — the number every ROI calculation in Week 3 needs.

4

What happens at LOE?

Model the cliff: generic entry at year X takes 80–90% of volume in 12 months for a small molecule; biosimilar erosion is slower. Promotion should stop before then.

5

Which stakeholder moves each line?

Payers/PBMs → GTN. HCPs → volume. Patients → abandonment, adherence, copay spend. Policymakers → Medicaid rebates, IRA negotiation, 340B. Map the P&L to the stakeholder slide.

Discounts vs. rebates: the same net price, very different plumbing

Excerpt-based: IQVIA, “How Will a Rebate Model Impact Cash Flow in the 340B Drug Pricing Program?” (2025)

Upfront discount

“With an upfront discount, the 340B hospital or clinic pays a reduced price at the time of purchase, lowering its initial acquisition cost.” The manufacturer gives up margin immediately and cannot verify patient eligibility before discounting — the source of “duplicate discount” disputes with Medicaid.

Rebate

“Under a rebate model, the covered entity initially pays a non-340B price at the time of purchase ... a rebate is then used to reduce the 340B purchase price to the 340B price.” The manufacturer sees claim-level data before paying — better integrity, but the provider carries the float.

- ▶ **The empirical question:** does the float matter? IQVIA's cash-flow models found financing costs under a rebate model “less than half of one percent” of the program's value — 0.19% for entity-owned pharmacies, 0.03% for contract pharmacies; even under unfavorable assumptions, under 1.2%.
- ▶ **The policy stakes:** HRSA approved a rebate-model pilot for eight manufacturers on Oct 30, 2025, focused on the first 10 drugs selected for Medicare price negotiation. The program reached ~\$180–200B at list in 2025 — “approaching the size of Medicare Part D.”
- ▶ **Why it's on a Week 1 slide:** the same decision (discount now vs. rebate later) shows up in every commercial contract, and the analyst who can model cash flow, timing and data integrity together is rare.

05

Codes and acronyms

The identifiers that let a claims analyst join a diagnosis to a drug to a doctor to a plan.

Four identifiers you must recognize on sight

Full course glossary handed out tonight (≈100 terms)

ICD-10-CM

E11.65

Diagnosis code. E11 = type 2 diabetes; .65 = with hyperglycemia. ~70,000 codes; letter + 2 digits, then up to 4 more characters after the decimal. Appears on medical claims and EHR problem lists.

Analytics use: Cohort definition: “patients with ≥2 claims for E11.x”

NDC

0002-1434-80

National Drug Code, 10 or 11 digits in 3 segments: labeler – product – package. Identifies the exact drug, strength, form and package size. On every pharmacy claim; hospital drugs may be billed by HCPCS instead.

Analytics use: Map NDC → molecule → brand → market

NPI

1234567893

National Provider Identifier, 10 digits (Luhn check). Type 1 = individual (HCP); Type 2 = organization (HCO). Public registry (NPPES) gives specialty, address. On every claim as prescriber / rendering / billing provider.

Analytics use: Join claims to CRM targets; HCP segmentation

HCPCS / CPT

J1817 • 99214

Procedure and product codes on medical claims. CPT (Level I): services, e.g. 99214 office visit. HCPCS Level II J-codes: physician-administered drugs, e.g. J1817 insulin infusion pump supply. Q-codes, S-codes; G-codes for Medicare.

Analytics use: Buy-and-bill drug volume; infusion vs. oral markets

Twenty acronyms you'll hear in your first week on the job

HCP / HCO / IDN	Health-care professional / organization / integrated delivery network	KOL	Key opinion leader — influential physician-researcher
MSL	Medical science liaison — field medical, non-promotional	PBM	Pharmacy benefit manager
GPO	Group purchasing organization	WAC / AWP / ASP	Wholesale acquisition cost / average wholesale price / average sales price (Medicare Part B)
GTN	Gross-to-net (deductions from list to net revenue)	PA / step edit	Prior authorization / step therapy — utilization management
Tier	Formulary tier — determines patient copay	LOE	Loss of exclusivity — patent/regulatory expiry; generic entry
DTC	Direct-to-consumer promotion	NRx / TRx / NBRx	New / total / new-to-brand prescriptions
RWD / RWE	Real-world data / real-world evidence	HEOR	Health economics & outcomes research
MLR	Medical-legal-regulatory review of promotional content	CRM	Customer relationship management (Veeva CRM is the standard)
SFE	Sales-force effectiveness	NBA	Next best action — orchestration of HCP touchpoints
IRA / MFP	Inflation Reduction Act / maximum fair price (Medicare negotiation)	340B	Federal drug-discount program for safety-net providers

06

Product lifecycle and launch archetypes

The questions a commercial analyst answers change completely every few years of a brand's life.

Launch → in-line → LOE: the questions change, and so does the data

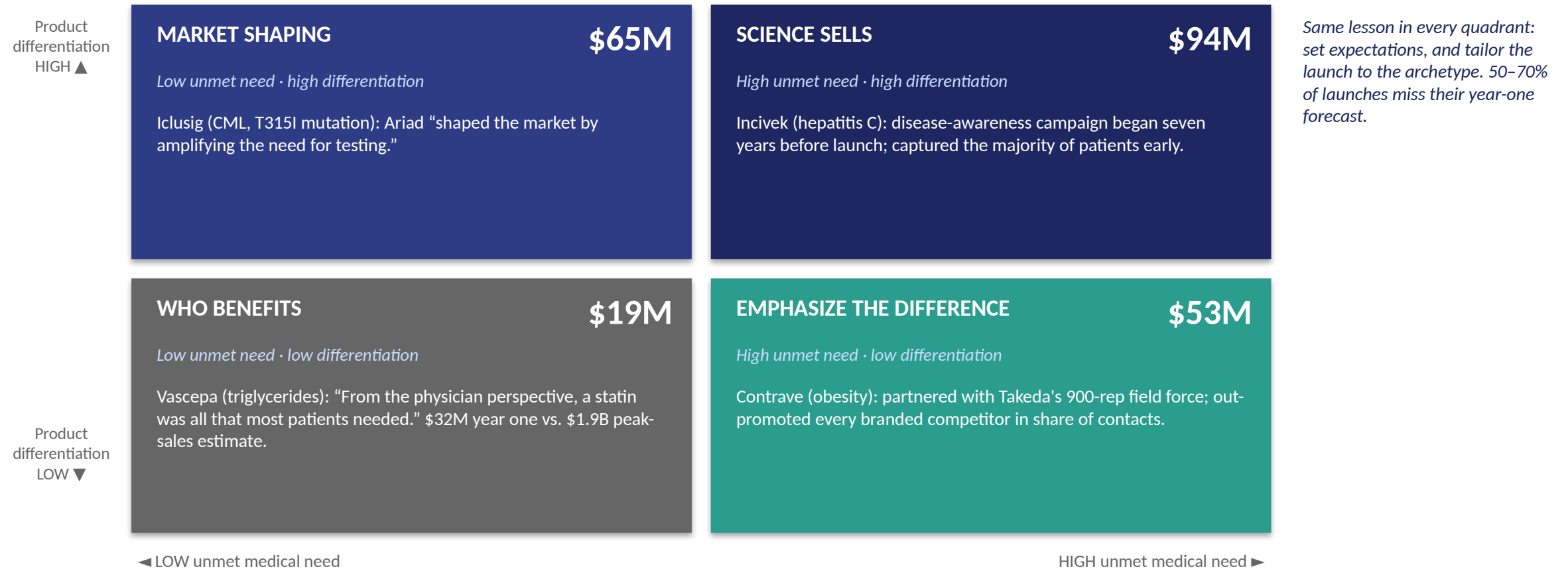
Concept from the PMSA webinar “Patient Data Applications Across Product Lifecycle” (course PMSA reading list)

Pre-launch (-3 to 0 yrs)	Launch (0-2 yrs)	In-line / growth (2-8 yrs)	LOE & beyond
▶ Market sizing: epidemiology → diagnosed → treated → addressable ▶ Analog selection; forecast the launch curve ▶ Payer landscape & pricing research ▶ Disease awareness (Incivek's 7-year runway)	▶ Weekly NBRx and writer tracking vs. forecast ▶ Early-adopter HCP targeting; sample and rep deployment ▶ Access wins/losses by payer; abandonment ▶ Rapid MMM recalibration	▶ Segmentation & NBA at scale ▶ MMM / ROI and budget reallocation ▶ Adherence, persistence, patient support ▶ Lifecycle: new indications, formulations	▶ Erosion forecast; generic/biosimilar entry timing ▶ Promotion wind-down; authorized generic strategy ▶ Contracting to defend covered lives ▶ Portfolio reallocation to the next launch

Where the course lives: Weeks 3-5 are mostly in-line questions; Week 6 (forecasting) spans all four phases; Week 2 (claims) is the raw material for everything.

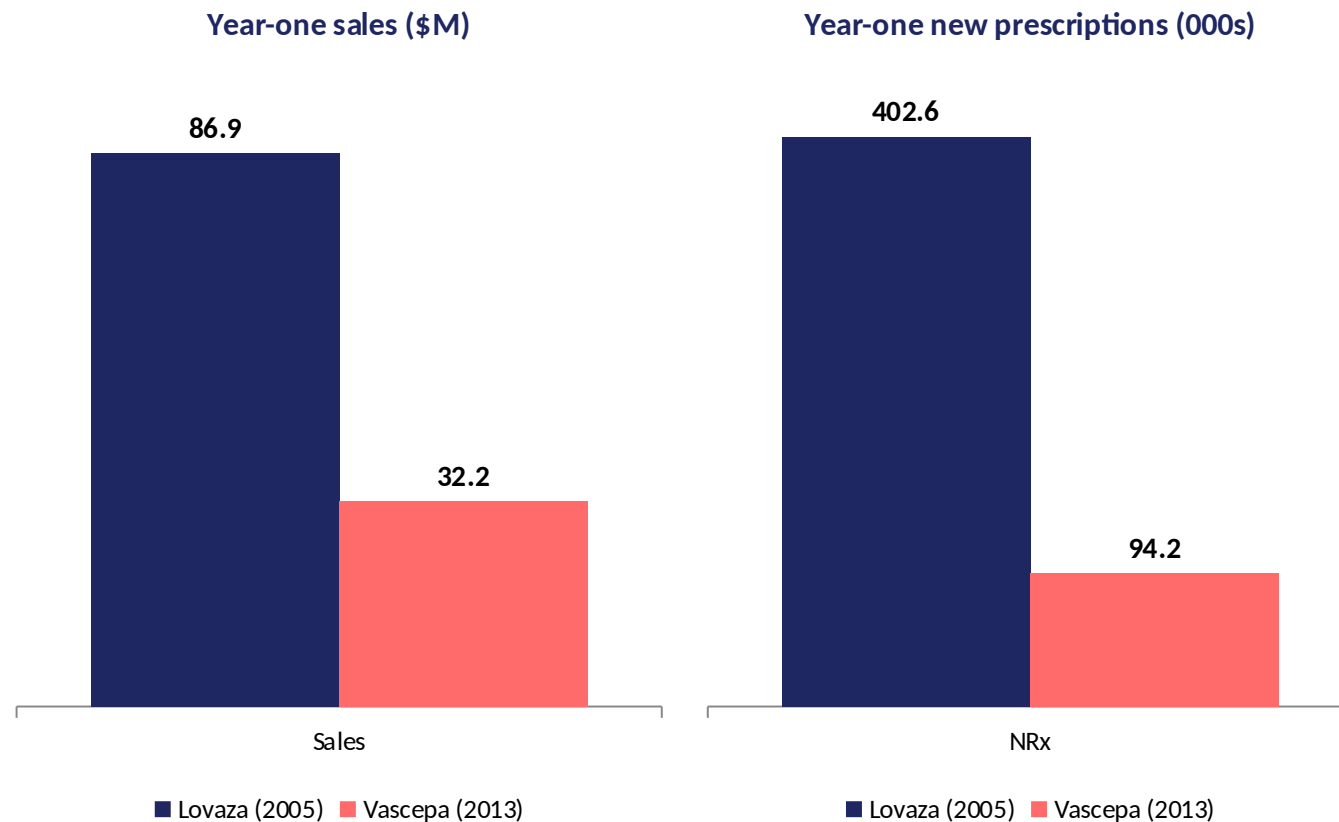
Four launch archetypes: unmet need × differentiation predicts year-one sales

Source: IQVIA, “A Framework for Successful Biopharma Launches” — 602 launches, 2007–2016; emerging-biopharma first-year sales shown



Two fish-oil launches, same market, 3× different outcome

IQVIA NSP / NPA, year-one performance. Lovaza launched Sep 2005; Vascepa Jan 2013 into a market where a statin “was all most patients needed.”



- ▶ **Same molecule class**, same indication (severely high triglycerides), same prescriber base.
- ▶ **Analyst forecasts:** \$59M in 2013 revenue and \$1.9B peak sales by 2022 — “inspired by the mechanism of action.”
- ▶ **Actual:** \$32.2M and 94.2k NRx vs Lovaza's \$86.9M and 402.6k.
- ▶ **Diagnosis (IQVIA):** “If marketers can't identify the distinct patient population or stakeholder who would benefit, the launch will likely fail.”
- ▶ **Your job in Week 6:** build forecasts that encode the stakeholder reality — payer coverage and HCP conviction — not just the mechanism.

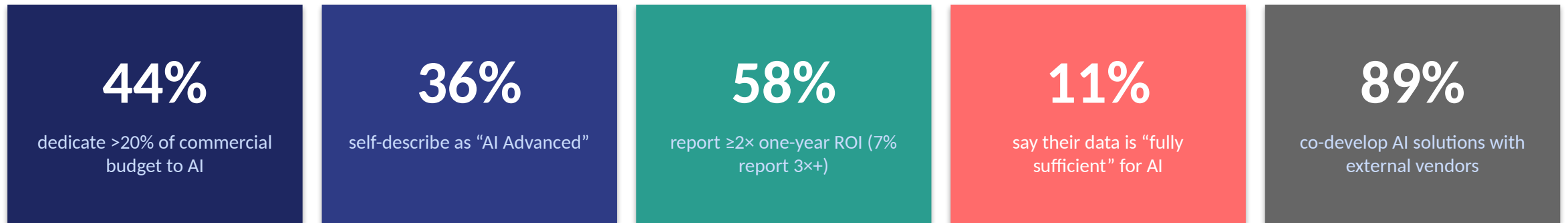
07

AI and agentic AI in commercial — why now

What 107 senior commercial leaders told IQVIA in 2025, and where GenAI projects actually fail.

AI has moved from pilots to budget line — but data readiness lags

Source: IQVIA, “AI in Life Sciences Commercialization” (May 2025 survey, n=107 senior commercial leaders)



Top barriers to scaling AI (% of respondents)



“We're not short on data — we're short on usable, connected data.”

“The ambition is there, but the infrastructure isn't.”

— survey respondents, IQVIA 2025. Translation for this course: the scarce skill is not fitting the model; it is building the cohort, joining the codes and knowing which stakeholder the number describes.

AI priority vs. adoption by function — and where this course maps

Adapted from IQVIA 2025 survey (Figure 5) and its priority use-case list

Function	Survey finding	Typical AI use cases	Course weeks
Sales & marketing	Highest strategic priority, low adoption — “barriers are about scaling enablement across decentralized teams”	Omnichannel targeting, NBA, content personalization, field productivity	Wk 3, 4, 5
Value & access	Overlooked opportunity — priority but even lower adoption	Payer analytics, pricing simulation, contract scenario modeling	Wk 2, 6
Compliance	Overlooked opportunity	Audit automation, MLR review assistance	Wk 7 (NLP)
Medical affairs	Moderate priority; human judgement remains the differentiator (“MSLs deliver scientific judgement; AI delivers information”)	Insight extraction from field notes, KOL identification, literature synthesis	Wk 7
IT & data operations	Leading in adoption — data infrastructure and integration	Data integration, master data (HCP/NDC), model deployment & monitoring	Wk 2, term project
Forecasting & supply	Top-cited current use case; moving to real-time adjustment	Demand forecasting, foundation models for time series	Wk 6

Where GenAI and agents pay off — and the four reasons projects fail

Source: IQVIA, “Where to Deploy Generative AI Solutions in Life Sciences” (2025). >70% of GenAI experiments never reach production.

Four reasons GenAI applications fail (IQVIA)

- ▶ **Unrealistic expectations:** “LLMs are unable to create insights if the information necessary for those insights isn't somewhere in the data sources.”
- ▶ **Doesn't fit the workflow:** “Inconvenience kills value.” Switching apps and slow responses hammer adoption.
- ▶ **Can't prove accuracy:** errors are less predictable than in deterministic software; measurement itself is subjective.
- ▶ **Not useful to expert users:** experts are already fast; the bar for improvement is high.

Rules of thumb for high-value use cases

- ▶ **Repetitive manual tasks:** “LLMs can rarely do things a person can't do if given enough time” — most wins automate human-like tasks.
- ▶ **Large volumes of text or code:** field notes, call transcripts, medical literature, MLR review, analytics code.
- ▶ **Intrinsic P&L impact:** an M&A screening tool used rarely can beat a slide-template generator used daily.
- ▶ **Human review + existing workflow + non-expert users:** lower risk of failure.
- ▶ **Agentic AI:** “empowered to perform dedicated tasks autonomously” — an orchestrator routes requests to specialist agents and synthesizes the answer (IQVIA's own design).

Course thread: every technique from Week 3 on ends with “what would the agentic version of this look like?” Guest Speaker 1 (Nov 16) covers agentic use cases in commercial.

08

Use cases, claims tour, and exercise

Fifteen starting points, one dataset to look at, and one exercise to do before you leave.

Term-project use cases — data science track

Five archetypes from the syllabus plus extensions. Full one-page briefs in the catalog handout.

#	Use case	Core question	Data (synthetic, provided or generated)	Methods
1	HCP segmentation + NBA engine	Which HCPs, which channel, which message, how often?	Rx panel + CRM activity	Clustering, uplift, bandits
2	MMM calibrated to a launch curve	What did each channel contribute, and how should we reallocate?	Weekly TRx + channel spend	Bayesian MMM, adstock/Hill
3	Patient-finding / journey model	Which undiagnosed or untreated patients look like our treated cohort?	Medical + pharmacy claims	Sequence models, survival
4	Adherence & persistence prediction	Who drops off therapy at 90 days, and what intervention helps?	Pharmacy claims + support-program data	Classification, survival, causal
5	Payer access impact analysis	What did a formulary change do to NBRx, abandonment and net price?	Claims with plan IDs + formulary events	Event study, diff-in-diff
6	Launch forecast with analogs	Epi funnel → patient flow → TRx; range forecast for year 1–3	Epi tables + analog launch curves	Bass diffusion, hierarchical TS
7	LOE erosion model	How fast does volume and net price erode after generic entry?	Historical analog brands	Time series, scenario analysis
8	KOL / influence network	Which physicians shape prescribing in a specialty and region?	Publications, referrals, claims co-management	Graph analytics

Term-project use cases — NLP and agentic track

Aim for a working prototype plus an honest evaluation, not a demo. Mentors will push on “would a brand team trust this?”

#	Use case	Core question	Data	Methods
9	Field-note insight extraction	What are reps and MSLs hearing about access, efficacy, competitors — by region and segment?	Synthetic CRM call notes	Embeddings, topic models, LLM extraction
10	Social-listening & adverse-event triage	Which posts describe a possible adverse event vs. a complaint vs. noise?	Synthetic social posts	Classification, domain models (BioBERT)
11	MLR pre-review assistant	Does this promotional claim match the label and the reference deck?	Label text + claim library	RAG, claim-matching, evaluation harness
12	Weekly brand-performance briefing agent	Automate the Monday brand report: pull data, detect anomalies, draft narrative, flag actions	Weekly TRx/share/access tables	Agent orchestration, tool use, evals
13	Payer-policy monitoring agent	Track formulary and policy documents; summarize changes and quantify exposure	Public formulary PDFs, policy pages	Document parsing, RAG, change detection
14	Synthetic advisory board	Can LLM personas grounded in real interview transcripts predict what an ad board would say?	Synthetic transcripts	Persona modeling, evaluation vs. holdout
15	Forecast-explanation agent	Explain forecast variance to a brand lead in plain language, with drivers and confidence	Forecast + actuals + drivers	Time-series decomposition + LLM narrative

Pick a problem a real brand team has. Then ask which stakeholder's behavior you are modeling — that decides your data and your evaluation metric.

Tour of an anonymized claims sample: what a row actually looks like

Notebook: Week_01_Claims_Sample_Tour.ipynb • 60 synthetic patients on GLUCOREN and competitors • 10 minutes, no coding required

Column (pharmacy claim)	Example	Which stakeholder / concept it encodes
patient_id (tokenized)	P00017	Patient — de-identified; HIPAA; longitudinal linkage across claims
service_date • days_supply	2026-03-04 • 30	Patient behavior: refills, gaps, persistence
ndc • drug_name	0002-1434-80 • GLUCOREN 10 mg	Manufacturer / product; NDC → molecule → market
prescriber_npi • specialty	1234567893 • Endocrinology	HCP — the chooser; join to CRM targets
plan_id • payer_type • pbm	PL-COM-04 • Commercial • PBM-A	Payer / PBM — the access gate
claim_status	PAID • REJECTED • REVERSED	Adjudication: rejections often = prior auth; reversals = abandonment
reject_code	75 (prior authorization required)	Utilization management in action
patient_pay • plan_pay • copay_card	\$45 • \$410 • \$30	Money flow: who paid what at the counter
pharmacy_type	Retail • Mail • Specialty	Channel; PBM-owned mail/specialty steering

Next week we build a real cohort (DLBCL, 500 patients, medical + pharmacy claims) and assign lines of therapy. Tonight is only about recognizing the fields.

Exercise: map the stakeholders for GLUCOREN (25 min, groups of 3–4)

Handout: Week_01_InClass_Exercise. Debrief in the last 15 minutes; one group presents each part.

A Part A • Incentive map (10 min)

- ▶ GLUCOREN: branded once-daily oral for type 2 diabetes, year 3, \$650M net sales, facing a new competitor with a cardiovascular-outcomes label
- ▶ For each of 8 stakeholders: what do they want from GLUCOREN, and what would make them act against it?
- ▶ Identify the two most important conflicts

B Part B • Gross-to-net (8 min)

- ▶ From the worksheet's WAC, units, rebate rates and channel mix, compute gross sales, each deduction, net sales, GTN %, and net revenue per TRx
- ▶ Then: the PBM asks for 5 more points of rebate for preferred status. Volume rises 12%. Is it worth it?

C Part C • Follow the prescription (7 min)

- ▶ A 58-year-old patient on a high-deductible plan gets a GLUCOREN script at an endocrinology visit
- ▶ Trace the 6 steps: which code appears where, which dataset captures it, what could stop the fill at each step
- ▶ Bonus: which of the term-project use cases would help at each failure point?

The answer key is in the instructor package; students get the worked solution after the debrief.

Before Week 2 (Sep 21): readings, teams, and one question to bring

Week 2: Pharma Datasets & Introduction to Claims Analytics — medical vs. pharmacy claims, EHR, syndicated panels; building a cohort with codes; RWD quality and re-identification risk.

Do

- ▶ **Read:** IQVIA Institute, “The Use of Medicines in the U.S.” — latest annual report, executive summary (free at iqvia.com/institute).
- ▶ **Read:** Week 1 reading pack (curated white-paper excerpts, ~30 min) and skim the glossary — the acronyms on tonight's slides will be assumed from now on.
- ▶ **Optional:** FDA, “Framework for FDA's Real-World Evidence Program” — executive summary and Section II (fda.gov/media/120060/download).
- ▶ **Form your team:** 3–4 people; post team name and members on the course site by Friday. Mentors are introduced next Monday.
- ▶ **Set up:** Python 3.11+, pandas, DuckDB, Jupyter. Clone the Week 2 data folder when posted.

Bring

- ▶ **One question** about a stakeholder incentive you found surprising tonight — we start Week 2 with five of them.
- ▶ **A one-sentence draft** of a commercial problem your team might want to work on (it will change; that's fine).
- ▶ **For guest speakers later in the term:** questions are submitted in advance and graded — start noticing what you'd want to ask a practitioner.

Office hours Mondays 5:00–6:45 PM • ac5562@columbia.edu